

Kaden Abzal

Lab Report: Lab 01 — The Price of One Request

1. Prediction and Measured Token Values (Part 1)

- Prediction: It was predicted that the English text would be the shortest, Russian would take an intermediate position, and Kazakh would require the highest number of tokens due to its complex agglutinative structure and specific characters (ә, ғ, қ, Һ, ө, ұ, Ү, Һ, і).
- Measured Values (Input Tokens, measured via Anthropic API on claude-opus-5):
 - English (EN): 145 tokens
 - Russian (RU): 209 tokens (1.44× of EN)
 - Kazakh (KK): 317 tokens (2.19× of EN)

2. Annual Costs at Estimated Volume

- Volume Justification: A volume of 2,000 requests per day is chosen as a realistic daily load metric for a mid-sized retail bank customer support center.

Model	English (USD/year)	Russian (USD/year)	Kazakh (USD/year)
Haiku-4.5	\$2,770	\$3,952	\$4,454
Sonnet-5	\$5,541	\$7,904	\$8,909
Opus-5	\$13,852	\$19,761	\$22,272
Fable-5.1	\$27,704	\$39,522	\$44,545

Note on methodology: Both input and output tokens were genuinely measured by sending one real request per language to claude-opus-5 via the Anthropic API (Part 2 of the lab, run in Google Colab). Output token counts were as billed: EN=730, RU=1041, KK=1157. The total-bill ratio ($1.00\times / 1.43\times / 1.61\times$) differs from the input-only ratio ($1.00\times / 1.44\times / 2.19\times$) because output length does not scale with input length in the same proportion across languages.

3. Model Selection for the Kazakh-Language Support Queue

For production in the Kazakh-language support queue, it is optimal to deploy the Sonnet-5 model (or Haiku-4.5 under strict budget constraints): it provides an optimal balance between manageable costs for processing increased token counts and high quality in following complex system prompt instructions. This is supported by direct observation during Part 2: the model's actual answers in all three languages correctly declined to fabricate a reason for the rate change and did not invent any account number, rate, or date not present in the complaint — exactly the behavior the system prompt required, minimizing the risk of hallucination on a low-resource language such as Kazakh.

4. Cost-Reduction Lever

This lab session did not utilize Prompt Caching, which significantly reduces expenses for repeatedly sending invariant system prompts across high volumes of incoming requests.

AI-Use Declaration

- Tools Used: Claude (Anthropic) and Gemini (Google), used for different purposes.
- Purposes of Use:
 1. Claude was used to understand the task requirements, troubleshoot the local environment (Anaconda terminal, GitHub fork and upload workflow), and clarify the distinction between a Claude Pro subscription and API console access.
 2. Since the instructor's Anthropic API key was initially unavailable, Claude helped identify Google Gemini's free API tier as a temporary alternative for an early attempt at Part 2, and reviewed that attempt's output — identifying that output tokens had been assumed rather than measured in that first version.
 3. Gemini was used to write the initial (Gemini-based) token-counting script in Google Colab, before an Anthropic API key was obtained.
 4. Once an Anthropic API key was obtained, Claude provided the official Part 2 measurement script (adapted for Google Colab from the course repository's `part2_measure.py`) and helped structure and format this final report using the resulting real measurements.